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$$b) A \cap C = B \cap C$$

Solution: $A = \{1, 2, 3, 9\}$ $B = \{1, 2, 3, 10\}$ $C = \{1, 2, 3, 4\}$

19) $A \cap C = B \cap C = \{1, 2, 3\}$. But $A \neq B$. $\square$

c) $A \cup C = B \cup C$ and $A \cap C = B \cap C$.

Solution: $\therefore (A \cup C) \cap (A \cap C) = (B \cup C) \cap (B \cap C)$

$$(A \cup C) \cap (A \cap C)$$

$\therefore A \cup C \subseteq B \cup C$ and $B \cup C \subseteq A \cup C$. Also $A \cap C \subseteq B \cap C$ and $B \cap C \subseteq A \cap C$.
Let, x be an arbitrary element of $A \cup C$, i.e. $x \in A \cup C$. By Defn. 3 from Section 2.1, $x \in B \cup C$.

By Defn. of ^{union}, $x \in A \vee x \in C$. By Defn. of ^{union}, $x \in B \vee x \in C$.

Suppose $x \notin C$. $\therefore$ By Defn. of not belongs to, $\neg(x \in C)$. By Disjunctive syllogism, $x \in A$ and $x \in B$ both holds.

$\therefore$ By Defn. of conditional, $x \notin C \rightarrow x \in A \wedge x \in B$.

Let, y be an arbitrary element of $A \cap C$, i.e. $y \in A \cap C$. By Defn. 3 from Section 2.1, $y \in B \cap C$. By Defn. 2, $y \in A \wedge y \in C$. By Defn. 2, $y \in B \wedge y \in C$.

By Defn. of AND, $y \in A$ and $y \in C$. By Defn. of AND, $y \in B$ and $y \in C$.

Let, p be an arbitrary element of A , i.e. $p \in A$.

We know that an element can either be a member of C or not a member of C .

Let, $p \notin C$. By Modus ponens, $p \in A \wedge p \in B$. By defn. of AND, $p \in B$.